higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

PLANT AND EQUIPMENT
NQF LEVEL 2

8 MARCH 2019

This marking guideline consists of 6 pages.

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QUESTION 1

1.1
1.1.1 True
1.1.2 False
1.1.3 True
1.1.4 True
1.1.5 True

(5 × 1) **[5]**

QUESTION 2

2.1
- It ensures the financial, physical and emotional wellbeing of workers.
- This makes workers and the workplace more productive and safe.
- It increases the profitability of a work place due to less risk of injuries.
- This leads to greater job security for workers and increased salaries or wages.
- PPE is needed to reduce the risk in a working environment.
- By using PPE it makes the workplace a safer place and encourages people to work safely and responsibly. (Any 5 × 1) (5)

2.2 Occupational Health and Safety Act (1)

2.3
- The act covers workers who are involved with the planning, layout, construction, use, alteration, repair, maintenance or demolition of buildings.
- The act covers workers who use plant, machinery and other equipment. This includes tool such as electric drills and equipment's like scaffolds. (4)

2.4
- Being injured by projecting nails, splinters, wires and broken glass.
- Bumping against badly packed or badly placed materials.
- Tipping over lose objects that are lying around.
- Slipping on wet, greasy or dirty floors.
- Being injured by materials or tools falling or dropping from above.
- Having fires on side. (Any 5 × 1) (5)

[15]

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QUESTION 3

3.1
- Highest priority✓ – people who need immediate help and transport to medical help.✓
- Second priority✓ – people who can probably wait for up to an hour for medical help without a risk to their lives.✓
- Lowest priority✓ – people who can wait and receive transport last or those who evidently passed on.✓ (6)

3.2
- Always wear protective gloves before you touch an injured person, especially if there is any blood.
- Always wash your hands after touching an injured person.
- Minimise mouth-to-mouth contact.
- Wear a disposable face mask if possible.
- Be careful of sharp objects at the scene of an accident, especially those that have blood on them because they may infect you. (5)

3.3
- Dizziness
- Faintness
- Blurred vision
- Collapse
- Paleness
- Sweating and anxiety
- Pale/grey skin that turns bluish grey
- Bluish-purple lips, tongue, earlobes and fingernails
- Cold and clammy skin
- Shallow or irregular breathing or gasping for air
- Changes in levels of consciousness
- Weak, rapid pulse (sometimes the radial pulse may be absent) (Any 4 × 1) (4)

[15]

QUESTION 4

4.1
- Measuring tape
- Spirit level
- Screwdriver
- Steel square (Any 1 × 1) (1)

4.2 (Alternate Relevant answers accepted for 4.2.1 – 4.2.5)

4.2.1 Spirit level – is used to check whether the surface is horizontal (level) or vertical (plumb).

4.2.2 Plumb bob – to check if whether something is plumbed (vertical).

4.2.3 Caulking gun – is used to squirt sealant into small cracks.

4.2.4 Set square – is used to set squares in technical drawing to create lines at 90/60/45 degrees to other lines.

4.2.5 Screwdrivers – are used to tighten and loosen screws.

(5 × 1) (5)

4.3
- Time is wasted
- One can injure him/herself
- It can also ruin the job
- It can delay the completion of the job
- Machine becomes less effective
- Productivity is lowered at operation level (6)

4.4 (Alternate Relevant answers accepted for 4.2.1 – 4.2.5)

4.4.1
- Pipe vice
- Pipe cutter
- Deburring tool
- Water pump pliers
- Basin wrench
- Flexible drain cleaning rod
- Guillotine
- Tin snip
- Soldering iron (Any 3 × 1) (3)

4.4.2
Marking gauge
Mitre box
Bench hook
Hand saw
Tenon saw
Coping saw
Pad saw
Compass saw
Brace and bit
Wheel brace
Smoothing plane
Chisels
Oil stone (Any 3 × 1) (3)

4.4.3
- Bricklayers trowel
- Pointing trowel
- Gauging trowel
- Industrial wheelbarrow
- Four pound hammer
- Raking tool
- V-joint tool
- Flat chisel
- Bolster chisel
- Joint rulers (Any 3 × 1) (3)

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4.4.4
- Tile cutter
- Notched trowel
- Nibbler
- Carbide scriber
- Masonry rubbing stone
- Rubber mallet
- Squeegee
- Sponge
- Tile spacers (Any 3 × 1) (3)

4.5
- Scribe a circle on the steel sheet using a divider.
- Keep the divider at the same opening span and mark off six equal spacing's on the circle circumference.
- Join the marks on the circumference with straight scribed lines. (3)

4.6
- Mortise chisel
- Cold chisel
- Bevel-edged chisel
- Butt chisels
- Carving chisels (Any 3 × 1) (3)

[30]

QUESTION 5

5.1
- Stepladders/two-legged ladder
- Straight ladders
- Extension ladders (Any 2 × 1) (2)

5.2
- The ladder can move sideways along the wall, which can cause your position on the ladder to become insecure.
- The ladder can topple over sideways which will cause both you and the ladder to fall onto the ground.
- You can lose your balance and accidentally push against the wall or pole that the ladder is learning against, causing the ladder to fall over backwards.
- You can lose your balance and in an attempt to regain your balance you may drop the tool you are holding, and this tool can fall onto someone on the ground and injure them.
- You can lose your balance and fall off the ladder. (Any 4 × 1) (4)

5.3
- Foot rubbers
- Hinges
- Gravity stays
- Joints
- Bracing
- Styles and rivets (4 × 1) (4)

[10]

QUESTION 6

6.1
- Knowledge of the applications and limitations of different types of scaffolding.
- Knowledge of the applications and safe working loads of scaffolding components.
- Knowledge of the classification for scaffolding platforms and the loading limitations for scaffolds with multi-level platforms.
- Knowledge of the requirements for stability against overturning of free-standing platforms and of the purpose of scaffolding ties.
- Documentation relevant to the inspector's role, including the scaffold inspection register and handover certificates. (5)

6.2
- Have a width of at least 1.2 m at the base.
- Be fitted with castors.
- Be secured to the structure if the width of the base is less than that given in above or the height is greater than the permissible height.
- Be provided with plan bracing at the base and at every alternate higher lift.
- Be provided with a ladder access. (5)

6.3 3,5✓:1✓
(Accept scaling of ratio, e.g.: 7:2 or 10,5:3) (2)

6.4 The most important point to consider is that the ladder access must be inside✓ the tower to prevent the tower from overturning.✓ (2)

6.5
- Make sure the hinge, board and stays are intact.
- The ground must be solid and level.
- It must be set to the same height.
- Never space the trestles more than 1,3m apart.
- The board must be placed comfortably on the trestle. (5)

6.6
- Bracing shall be provided to prevent distortion of scaffolding.
- Bracing shall be arranged in triangular shaped patterns with connections at a distance not exceeding 300mm from the intersections of vertical and horizontal members.
- Longitudinal bracing on the outside of putlog and independent scaffolding shall be fixed at spacing not exceeding 10 bays horizontally and to the full height of the scaffolding.
- Bracing for ledgers shall be fixed to the full height of the scaffolding, and ledger to ledger.
- The bracing shall be fixed by means of brace couplers or double couplers near alternate pairs of standards.
- Plan bracing between the inner and outer rows of standards shall be placed across the diagonal of a bay and shall be provided at spacing's not exceeding 10 bays horizontally and, where possible at the tie levels.

(Any 5 × 1) (5)

6.7 Safe Working Load. (1)

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TOTAL: 100

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